API Gateway: Endpoint Reference

Flash Sale Purchase Platform. Single entry point for web and mobile clients.

1. Gateway responsibilities

Authentication of every request (token checked before routing). **Rate limiting** per user, then admission through the Traffic Controller (queue, token bucket, dynamic throttling). **Request routing** to the core microservices over REST. All traffic over **HTTPS**, after CDN/WAF and the load balancer. Each request carries a correlation ID, and purchase calls carry an `Idempotency-Key` header.

2. Endpoints

Method	Endpoint	Routed to	Purpose
POST	`/flash-sale/buy *`	Reservation (via Inventory, Payment, Order)	Start a purchase: stock check, reservation, payment, order
GET	`/flash-sales/{id}`	Product / Flash Sale	Sale status, start/end, remaining units
GET	`/products`	Product	List and browse products
GET	`/products/{id}`	Product	Product details and price
GET	`/products/search`	Search	Search with facets and suggestions
GET	`/products/{id}/availability`	Inventory	Check stock availability
GET	`/cart`	Cart	View cart and total
POST	`/cart/items`	Cart	Add item to cart
PUT	`/cart/items/{id}`	Cart	Update item quantity
DELETE	`/cart/items/{id}`	Cart	Remove item from cart
GET	`/reservations/{id}`	Reservation	Reservation status and expiry
POST	`/reservations/{id}/cancel`	Reservation	Cancel a reservation, release stock
GET	`/payments/{id}`	Payment	Payment status
POST	`/payments/{id}/refund`	Payment	Refund a payment (restricted)
POST	`/webhooks/payment`	Payment	Callback from Stripe / Razorpay (signed)
GET	`/orders`	Order	List the user's orders
GET	`/orders/{id}`	Order	Order details and shipping status
POST	`/orders/{id}/cancel`	Order	Cancel an order

***** Defined in the purchase-flow sequence diagram (`POST /flash-sale/buy`). The other endpoints are derived from the services and class methods in the design (catalog, cart, inventory, reservation, payment, order) and are proposed paths: rename or version them (for example `/api/v1/...`) to match your final contract.

3. Standard responses

200/201 success **202** accepted or queued **400** invalid request **401** not authenticated **409** sold out or duplicate request **410** reservation expired **429** rate limited **502/503** downstream unavailable. A successful purchase returns `txnId`, `reservationId` and `orderId`.